

Name	()	Class	
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南 华 中 学

NAN HUA HIGH SCHOOL

END OF YEAR EXAMINATION 2024

Subject : Mathematics (Part 2)
Paper :4052
Level : Secondary Two Express
Date : 7 October 2024
Duration : 1 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid / tape.

Answer **all** the questions. The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question, it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 60.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

FOR MARKER'S USE ONLY		
Question Number	Maximum Mark	Mark
1	2	
2	5	
3	9	
4	4	
5	8	
6	6	
7	7	
8	5	
9	3	
10	11	
Total Mark	60	

This paper consists of 18 printed pages.

Answer **all** questions

- 1 12 men take 4 days to make 9 cupboards. Assume that the rate of work done by each worker is constant.
How long will it take 4 men to make 24 cupboards?

Answer [2]
..

2 (a) Factorise completely $6x^2 + xy - y^2$.

Answer [2]

(b) Given that $\left(\frac{2}{5}x - y\right)^2 = 32$ and $xy = 10$, find the value of $4x^2 + 25y^2$.

Answer [3]

- 3 Mr Chew is a stationery shop owner who bought 150 notebooks at \$ x each and 180 pencils at \$ y each.

- (a) A festive season sale is approaching soon. To prepare for this sale, Mr Chew needs to pack these stationery equally into as many identical gift packs as possible, without any leftovers.

Find the maximum number of gift packs that he can pack.

Answer [2]

- (b) Mr Chew sold each gift pack for \$ $\left(2x + \frac{34}{9}y\right)$. He made a loss of \$312 from the sale. Form an equation and show that it simplifies to

$$90x + \frac{200}{3}y = 312. \quad [2]$$

Answer

- (c) It is given further that $2x : y = 56 : 9$.
Write down an equation relating x and y .

Answer

[1]

- (d) Hence, by solving the simultaneous equations, find the cost of 1 notebook and cost of 1 pencil.

Answer Cost of 1 Notebook is \$.....

Cost of 1 Pencil is \$..... [3]

- (e) Explain algebraically why Mr Chew can never make a profit from the sale.

Answer

[1]

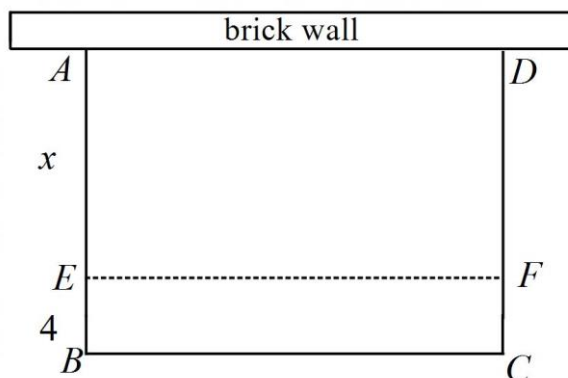
- 4 (a) Make b the subject of the formula $\frac{1}{a} + \frac{2}{b} = \frac{3}{ac}$.

Answer [3]

- (b) Given that $a > 0$ and $c > 0$, James claims that if he wants to obtain a positive value of b , $c > 0$ is a sufficient condition.
Explain whether James is correct, with appropriate working.

Answer [1]

5 (a)



A farmer wants to build an enclosure having two rectangular stalls, side by side. He constructs a brick wall on one side and uses 108 metres of wire fencing to construct the other sides of the enclosure, $AEBCFD$. Given that $AE = x$ metres and $EB = 4$ metres, show that the total area, A , of the enclosure is given by $A = 400 + 92x - 2x^2$.

Answer

[2]

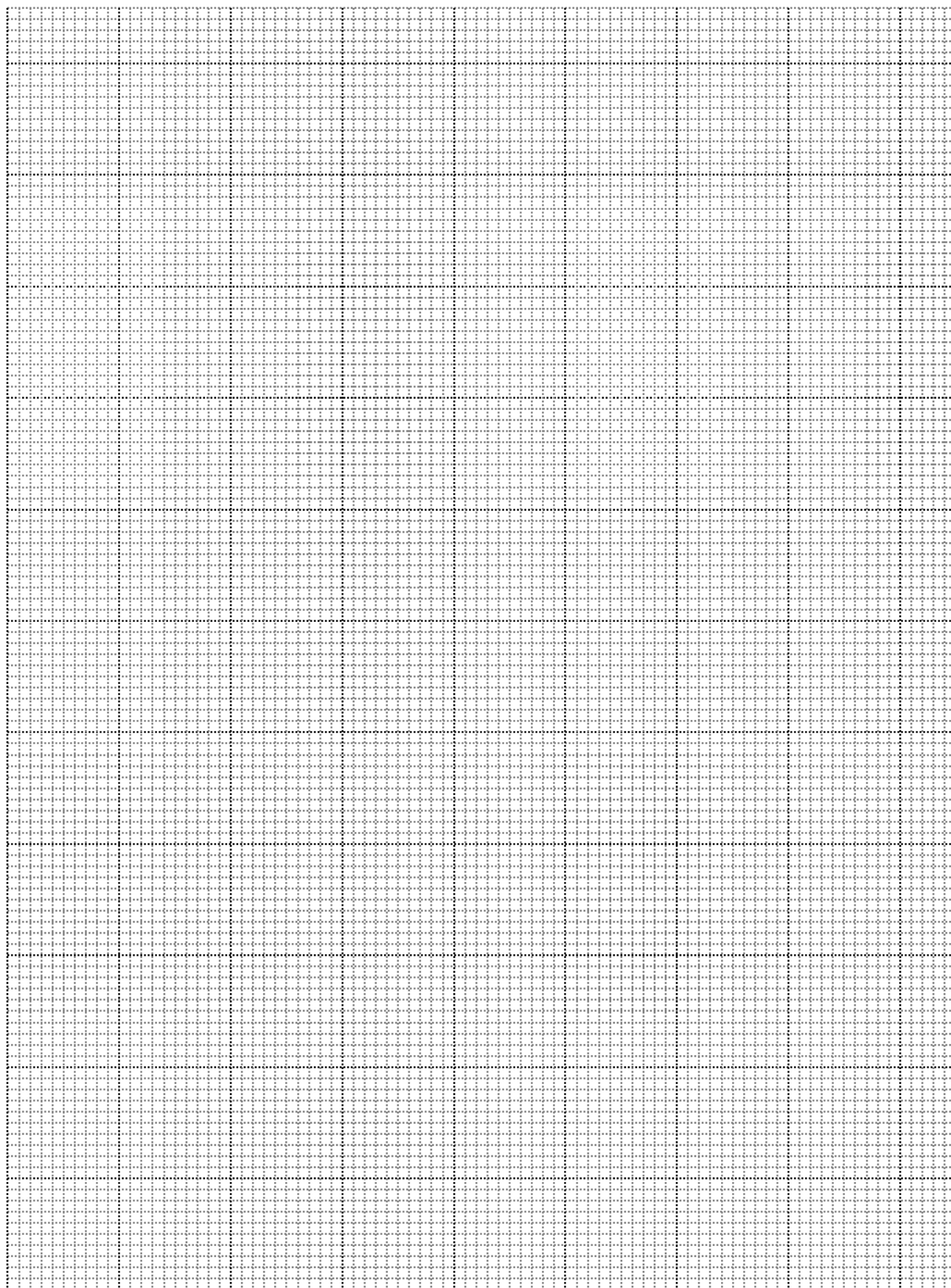
- (b) Using a scale of 2 cm to represent 10 metres on the horizontal x -axis and 2 cm to represent 200 metre^2 on the vertical A -axis, draw the graph of $A = 400 + 92x - 2x^2$ for $0 \leq x \leq 50$ on the grid provided on the next page.

x	0	10	20	30	40	50
A	400	1120	1440	1360	880	0

[3]

- (c) Given that the side AE is 35 m, use your graph to find the area of the enclosure.

Answer m^2 [1]



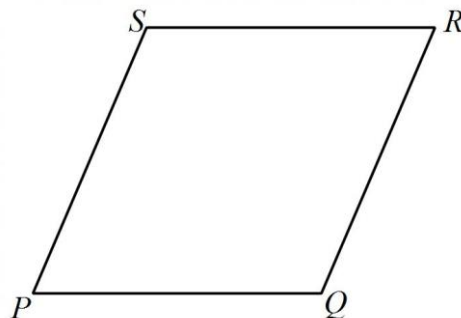
- 5 (d) The farmer wishes for his farm to be as large as possible. Using your graph, describe how you would advise him to find the dimensions of the enclosure.

Answer

[2]

- 6 $PQRS$ is a rhombus with a perimeter of 60 cm. The longer diagonal, PR is 24 cm.

- (a) Find $\angle QPR$.



Answer [2]

- (b) The length of the sides SR and PQ are now increased by 3cm each.

- (i) Write down the name of the new shape formed.

Answer [1]

- (ii) Hence, using your answer from (a) and (b)(i), find the area of the new shape.

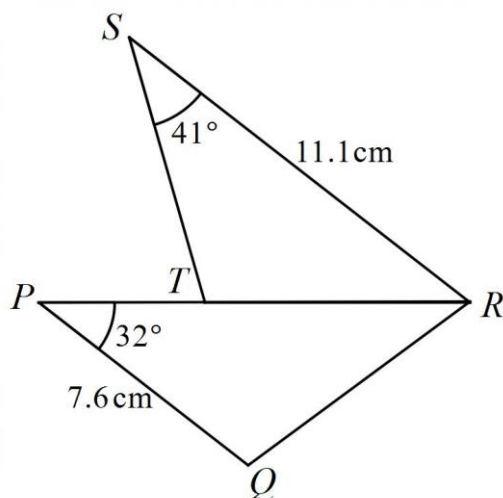
Answer cm^2 [3]

7 (a) Solve the equation $\frac{4x}{3x^2 - 13x + 12} + \frac{7}{3x - 4} = 0$.

(b) Express $\frac{n+2}{1-4n^2} - \frac{n-1}{3+6n}$ as a single fraction in its simplest form.

Answer $x = \dots\dots\dots$ [3]

- 8 In the diagram, $\triangle PQR$ and $\triangle RTS$ are congruent. $\angle QPR = 32^\circ$, $\angle RST = 41^\circ$, $PQ = 7.6$ cm and $RS = 11.1$ cm. [4]



(a) Find

(i) $\angle STR$,

Answer [2]

(ii) the length of PT .

Answercm [2]

(b) What can you conclude about lines PQ and SR ? Explain your reasoning clearly.

Answer

[1]

- 9 A class of 24 students took a mathematics test. The following stem-and-leaf diagram represents the scores of all the students.

Stem	Leaf
0	8 8 9
1	2 2 4 4 5
2	0 1 3 4 6 8 9
3	0 2 2 3 5 5
4	3 6 7

Key: 2 | 3 means 23 marks

- (a) Using the data above, find
(i) the median score,

Answer [1]

- (ii) the percentage of students who did not score more than 26 marks.

Answer% [1]

- (b) A student's score of 46 marks was recorded wrongly. His score should be 16 marks. Explain how it affects the median of the distribution.

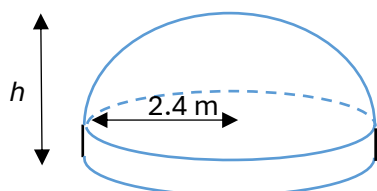
Answer [1]

- 10** Tent-makers Company manufactures custom-made tents. Customers can select the design of the tent as well as the type of fabric to use for manufacturing it.

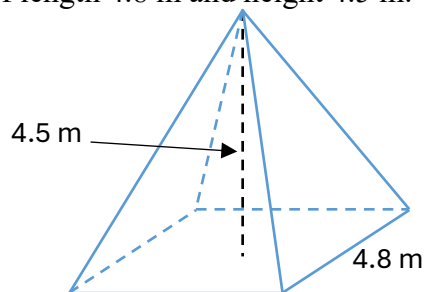
The two designs available are as follows:

Design *A* is modelled as a hemisphere of radius 2.4 m attached to a cylinder, and

Design *B* is modelled as a square pyramid of length 4.8 m and height 4.5 m.



Design A



Design B

- (a)** Assuming that the thickness of the tent can be ignored and that both tents contain the same volume when set up, show that $h = 2.71\text{m}$.

Answer

[3]

Table 1 below shows the range of masses for each type of fabric.

Fabric Type	Range of masses (in gsm*)
light-weight	30-200
medium-weight	200-350
heavy-weight	>350

*One gsm refers to the mass in grams of one square metre of material.

For either design, the whole tent including its base will be made of only one material to be chosen from Table 2 below.

Material code	Fabric Type	Water proof	U.V protection	Heat Insulation	Cost per square metre
1101	light-weight	Yes	Yes	Partially	\$4.30
1102	medium-weight	Yes	No	No	\$4.00
1103	medium-weight	Yes	Partially	Yes	\$4.45
1104	medium-weight	Yes	Partially	Yes	\$4.95

Mary is a recreational camper who wishes to purchase a tent from Tent-makers Company.

The following are her considerations when selecting the material for making the tent:

- The material must be of medium-weight type.
- The chosen design of the tent should make the tent as light as possible.
- She is on a tight budget and travelling to a place with heavy downpour, cold weather and minimal sunlight.

- (b) Based on the information above, which design (*A* or *B*) and which material (code 1101, 1102, 1103 or 1104) should Mary choose?
Explain your choices clearly.

Answer

Choose Design ____ because _____

_____.

Choose Material code _____ because _____

_____.

_____.

[6]

- (c) Mary claims that the mass of her tent will not exceed 26 kg. Explain whether her claim is correct.

Answer

[2]

End of Part 2

Name	()	Class	
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南 华 中 学

NAN HUA HIGH SCHOOL

END OF YEAR EXAMINATION 2024

Subject : Mathematics (Part 2) SOLUTIONS

Paper : 4052

Level : Secondary Two Express

Date : 7 October 2024

Duration : 1 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Answer **all** questions

FOR MARKER'S USE ONLY		
Question Number	Maximum Mark	Mark
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total Mark	60	

This paper consists of **13** printed pages.

- 1 12 men take 4 days to make 9 cupboards. Assume that the rate of work done by each worker is constant.
How long will it take 4 men to make 24 cupboards?

No of Men	No. of Cupboards	No of days
12	9	4
12	1	$\frac{4}{9}$
12	24	$\frac{96}{9}$
1	24	$\frac{96}{9} \times 12$
4	24	$\frac{96}{9} \times 12 \div 4$ $= 32$

- 2 (a) Factorise completely $6x^2 + xy - y^2$.

(a)	$6x^2 + xy - y^2 = (3x - y)(2x + y)$	
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- (b) Given that $\left(\frac{2}{5}x - y\right)^2 = 32$ and $xy = 10$, find the value of $4x^2 + 25y^2$.

(b)	$\left(\frac{2}{5}x - y\right)^2 = \frac{4}{25}x^2 - \frac{4}{5}xy + y^2$ $\frac{4}{25}x^2 - \frac{4}{5}xy + y^2 = 32$ $\frac{4}{25}x^2 - \frac{4}{5}(10) + y^2 = 32$ $\frac{4}{25}x^2 - 8 + y^2 = 32$ $\frac{4}{25}x^2 + y^2 = 32 + 8$ $4x^2 + 25y^2 = 1000$	
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- 3 Mr Chew is a stationery shop owner who bought 150 notebooks at \$ x each and 180 pencils at \$ y each.

- (i) A festive season sale is approaching soon. To prepare for this sale, Mr Chew needs to pack these stationery equally into as many identical gift packs as possible, without any leftovers.
Find the maximum number of gift packs that he can pack.

(i)	$150 = 2 \times 3 \times 5^2$ $180 = 2^2 \times 3^2 \times 5$ $\text{HCF} = 2 \times 3 \times 5$ $= 30$	\
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- (ii) Mr Chew sold each gift pack for \$ $\left(2x + \frac{34}{9}y\right)$. He made a loss of \$312 from the sale. Form an equation and show that it simplifies to $90x + \frac{200}{3}y = 312$.

(ii)	$150x + 180y - 30\left(2x + \frac{34y}{9}\right) = 312$ $150x + 180y - 60x - \frac{340}{3}y = 312$ $90x + \frac{200}{3}y = 312 \quad (\text{shown})$ -----(2)	
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- (iii) It is given further that $2x : y = 56 : 9$.
Write down an equation relating x and y .

(iii)	$\frac{2x}{y} = \frac{56}{9}$ $18x - 56y = 0 \quad \text{-----}(1)$ $9x - 28y = 0$	
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- (iv) Hence, by solving the simultaneous equations, find the cost of 1 notebook and cost of 1 pencil.

(iv)	$(1) \times 10:$ $90x - 280y = 0 \text{-----}(3)$ $(2)-(3):$ $346\frac{2}{3}y = 312$ $y = 0.9$ $x = 28(0.9) \div 9$ $= 2.8$ Cost of a notebook is \$2.80. Cost of a pencil is \$0.90.	
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- (v) Explain algebraically why Mr Chew can never make a profit from the sale.

Answer

$$\begin{aligned}
 \text{Profit} &= 30\left(2x + \frac{34y}{9}\right) - (150x + 180y) \\
 &= 60x + \frac{340}{3}y - 150x - 180y \\
 &= -90x - \frac{200}{3}y \\
 &= -(90x + \frac{200}{3}y)
 \end{aligned}$$

Since $x > 0$, $y > 0$, profit < 0 .

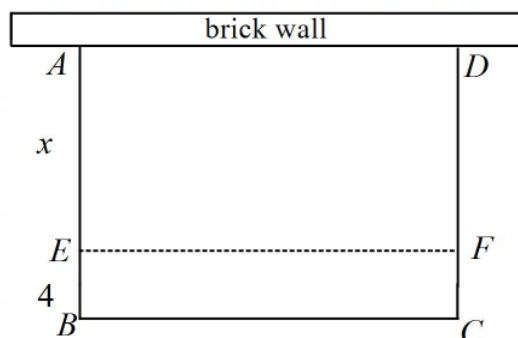
- 4 (a) Make b the subject of the formula $\frac{1}{a} + \frac{2}{b} = \frac{3}{ac}$.

(a)	$\frac{1}{a} + \frac{2}{b} = \frac{3}{ac}$ $bc + 2ac = 3b$ $bc - 3b = -2ac$ $b(c - 3) = -2ac$ $b = \frac{-2ac}{c - 3}$ $= \frac{2ac}{3 - c}$	
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- (b) Given that $a > 0$ and $c > 0$, James claims that if he wants to obtain a positive value of b , $c > 0$ is sufficient.
Explain whether James is correct with appropriate working.
Answer

(b)	It is not sufficient. $3 - c > 0$ $c < 3$ $0 < c < 3$	
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5 (a)



A farmer wants to build an enclosure having two rectangular stalls, side by side. He constructs a brick wall on one side and uses 108 metres of wire fencing to construct the other sides of the enclosure, $AEBCFD$. Given that $AE = x$ metres and $EB = 4$ metres, show that the total area, A , of the enclosure is given by $A = 400 + 92x - 2x^2$.

(a)	$BC = 108 - (x + 4 + x + 4)$ $= 108 - x - 4 - x - 4$ $= 100 - 2x$ $A = (x + 4)(100 - 2x)$ $= 100x - 2x^2 + 400 - 8x$ $= -2x^2 + 92x + 400$	
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(b) Using a scale of 2 cm to represent 10 metres on the horizontal x -axis and 2 cm to represent 200 metre² on the vertical A -axis, draw the graph of $A = 400 + 92x - 2x^2$ for $0 \leq x \leq 50$ on the grid provided on the next page.

x	0	10	20	30	40	50
A	400	1120	1440	1360	880	0

G1- correct scale and axes

G1-plot all points correctly

G1-smooth and symmetrical curve about turning point (22 to 24)(theoretical is (23, 1458)

(c) Given that the side AE is 35 m, use your graph to find the area of the enclosure.

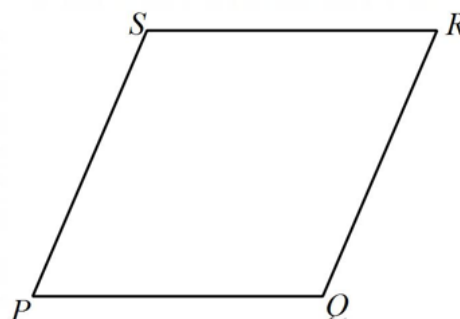
(c)	1170 $\pm$ 100	
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- 5 (d) The farmer wishes for his farm to be as large as possible. Using your graph, describe how you would advise him to find the dimensions of the enclosure.

(d)	Maximum point is (23, 1460). Maximum area occurs when $x = 23$. Dimensions are $(23+4)$ by $(100-2(23))$ $= 27$ m by 54 m OR Describe: look at maximum point Describe: its corresponding x-coordinate, find its new length by adding 4 to x and new breadth by $100-2x$	
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- 6 $PQRS$ is a rhombus with a perimeter of 60 cm. The longer diagonal, PR is 24 cm.

(a) Find $\angle QPR$.



(a)	$\cos \angle QPR = \frac{12}{15}$ $\angle QPR = \cos^{-1} \frac{12}{15}$ $= 36.8699^\circ$ $= 36.9^\circ (\text{to 1 d.p})$	
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- (b) The length of the sides SR and PQ are now increased by 3cm.

(i) Write down the name of the new shape formed.

(i)	parallelogram	
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(ii) Hence, using your answer from (a) and (b)(i), find the area of the new shape.

(ii)	$\angle SPQ = 2 \times 36.8699^\circ$ $= 73.7398^\circ$ Height of parallelogram: $\sin \angle SPQ = \frac{\text{height}}{15}$ $\text{height} = 15 \sin 73.7398^\circ$ $= 14.400$ Area of parallelogram = 18×14.400 $= 259 \text{ cm}^2 \text{ (3sf)}$	
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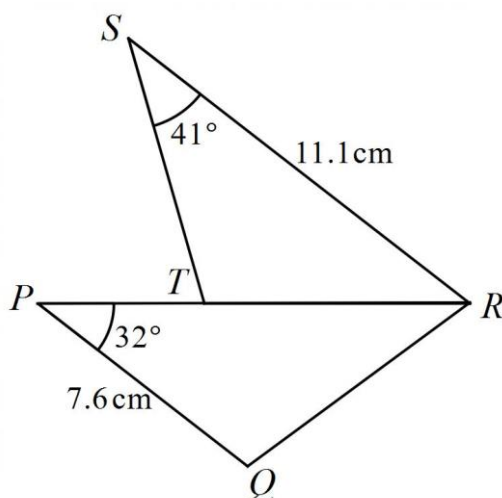
7 (a) Solve the equation $\frac{4x}{3x^2 - 13x + 12} + \frac{7}{3x - 4} = 0$.

(a)	$\frac{4x}{3x^2 - 13x + 12} + \frac{7}{3x - 4} = 0$ $\frac{4x}{(3x - 4)(x - 3)} + \frac{7}{3x - 4} = 0$ $4x + 7(x - 3) = 0$ $4x + 7x - 21 = 0$ $11x = 21$ $x = 1\frac{10}{11}$	
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(b) Express $\frac{n+2}{1-4n^2} - \frac{n-1}{3+6n}$ as a single fraction in its simplest form.

(b)	$\frac{n+2}{1-4n^2} - \frac{n-1}{3+6n} = \frac{n+2}{(1+2n)(1-2n)} - \frac{n-1}{3(1+2n)}$ $= \frac{3(n+2) - (n-1)(1-2n)}{3(1+2n)(1-2n)}$ $= \frac{3n+6 - (n-2n^2-1+2n)}{3(1+2n)(1-2n)}$ $= \frac{3n+6-n+2n^2+1-2n}{3(1+2n)(1-2n)}$ $= \frac{2n^2+7}{3(1+2n)(1-2n)}$	
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- 8 In the diagram, $\triangle PQR$ and $\triangle RTS$ are congruent. $\angle QPR = 32^\circ$, $\angle RST = 41^\circ$, $PQ = 7.6$ cm and $RS = 11.1$ cm.



(a) Find

(i) $\angle STR$,

(i)	$\angle SRT = \angle RPQ$ (corresponding $\angle$ s are equal) $= 32^\circ$ $\angle STR = 180^\circ - 32^\circ - 41^\circ$ ($\angle$ sum of Δ) $= 107^\circ$	
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(ii) the length of PT .

(ii)	$PR = RS$ (corresponding sides are equal) $= 11.1\text{ cm}$ $TR = QR$ (corresponding sides are equal) $= 7.6\text{ cm}$ $PT = 11.1 - 7.6$ $= 3.5\text{ cm}$
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- (b) What can you conclude about lines PQ and SR ? Explain your reasoning clearly.
Answer

(b)	Since $\angle SRT = \angle RPQ$, alternate angles are equal and $PQ \parallel SR$.	
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- 9 A class of 24 students took a mathematics test. The following stem-and-leaf diagram represents the scores of all the students.

Stem	Leaf
0	8 8 9
1	2 2 4 4 5
2	0 1 3 4 6 8 9
3	0 2 2 3 5 5
4	3 6 7

Key: 2 | 3 means 23 marks

- (a) Using the data above, find
(i) the median score,

(i)	$\text{median score} = \frac{24 + 26}{2}$ $= 25 \text{ marks}$	
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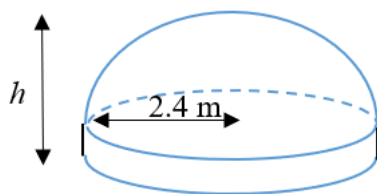
- (ii) the percentage of students who did not score more than 26 marks.

(ii)	$\% \text{ of students} = \frac{13}{24} \times 100\%$ $= 54 \frac{1}{6} \%$ $= 54.2\% \text{ (to 3s.f)}$	
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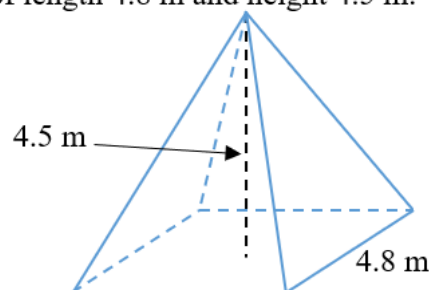
- (b) The student's score of 46 marks was recorded wrongly. His score should be 16 marks. Explain how it affects the median of the distribution.

(b)	Since correct score is less than original median score, there will be a decrease in the median score. (If student calculate the new median score to be 23.5, needs to still compare to the original 25 marks)	
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- 10 Tent-makers Company manufactures custom-made tents. Customers can select the design of the tent as well as the type of fabric to use for manufacturing it. The two designs available are as follows:
Design *A* is modelled as a hemisphere of radius 2.4 m attached to a cylinder, and Design *B* is modelled as a square pyramid of length 4.8 m and height 4.5 m.



Design *A*



Design *B*

- (a) Assuming that the thickness of the tent can be ignored and that both tents contain the same volume when set up, show that $h = 2.71\text{ m}$.

(a)	volume of pyramid = $\frac{1}{3}(4.8)^2(4.5)$ = 34.56 m^3 Let h_1 be the height of the cylinder. $\frac{2}{3}\pi(2.4)^3 + \pi(2.4)^2 h_1 = 34.56$ $h_1 = \frac{34.56 - \frac{2}{3}\pi(2.4)^3}{\pi(2.4)^2}$ = 0.30986 $h = 2.4 + 0.30986$ = 2.70986 = 2.71 m (shown)	
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Table 1 shows the range of masses for each type of fabric.

Fabric Type	Range of masses (in gsm*)
light-weight	30-200
medium-weight	200-350
heavy-weight	>350

Table 1

*One gsm refers to the mass in grams of one square metre of material.

For either design, the whole tent including its base will be made of only one material to be chosen from Table 2.



Material code	Fabric Type	Water proof	U.V protection	Heat Insulation	Cost per square metre
1101	light-weight	Yes	Yes	Partially	\$4.30
1102	medium-weight	Yes	No	No	\$4.00
1103	medium-weight	Yes	Partially	Yes	\$4.45
1104	medium-weight	Yes	Partially	Yes	\$4.95

Table 2

Mary is a recreational camper who wishes to purchase a tent from Tent-makers Company.

The following are her considerations when selecting the material for making the tent:

- The material must be of medium-weight type.
- The chosen design of the tent should make the tent as light as possible.
- She is on a tight budget and travelling to a place with heavy downpour, cold weather and minimal sunlight.

- (b) Based on the information above, which design (A or B) and which material (code 1101, 1102, 1103 or 1104) should Mary choose? Explain your choices clearly.

$$\begin{aligned} \text{surface area of hemi.+cylinder} &= 2\pi(2.4)^2 + 2\pi(2.4)(0.30986) + \pi(2.4)^2 \\ &= 58.9593m^2 \\ \text{slant height of pyramid} &= \sqrt{4.5^2 + 2.4^2} \\ &= \sqrt{26.01} \\ &= 5.1 \\ \text{surface area of pyramid} &= (4.8)(4.8) + 4 \times \left(\frac{1}{2} \times 4.8 \times 5.1\right) \\ &= 72m^2 \end{aligned}$$

I choose Design A because it has a smaller surface area.
I choose Material code 1103 because it is medium-weight, water proof, has heat insulation and is the lowest cost per square metre.

- (c) Mary claims that the mass of her tent will not exceed 26 kg. Explain whether her claim is correct.

(ii)	$\begin{aligned}\text{upper limit mass} &= 350 \times 58.9593 \\ &= 20635.959g \\ &= 20.6kg\end{aligned}$ Since the upper limit mass is 20.6kg, the mass won't exceed 26kg.	.
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End of Part 2